Exercise 24.1

Give an intuitive explanation why the estimator σ_N is biased, and how the bias depends on the number of observations.

Solution:

Exercise 24.2

Marginalize over σ and obtain the posterior marginal distribution of the mean μ given the data and a prior for μ and σ.

Solution:

Exercise 24.3

[This exercise requires macho integration capabilities.] Give the posterior predictive distribution for a new observation x_{N+1} given data $\{x_n\}$ and a prior on (μ, σ), and show how the result reduces to the Student-t distribution in the appropriate limit.

Solution: